


Group Homomorphisms

Def An isomorphism from $(G, *)$ to $(G', \circ)$

is a bijection $f: G \rightarrow G'$ such that

$$\underline{f(a * b) = f(a) \circ f(b)} \quad \text{operation preserving}$$

Def A homomorphism of groups, $(G, *)$ & $(G', \circ)$

is a function $f: G \rightarrow G'$ with

$$f(a * b) = f(a) \circ f(b).$$

Eg $GL_n(\mathbb{R}) = \{ n \times n \text{ real matrices w/ non-zero determinant} \}$
a group under matrix mult.

$\det: GL_n(\mathbb{R}) \rightarrow \mathbb{R}^* = \text{multiplicative group in } \mathbb{R}$

$$\det(A \cdot B) = \det(A) \cdot \det(B)$$

$$\ln: (\mathbb{R}^{>0}, \cdot) \rightarrow \mathbb{R}^+$$

positive reals
under mult.

$$\ln(x \cdot y) = \ln x + \ln y$$

Claim If $f: G \rightarrow H$ is a hom. then

1) $f(e_G) = e_H$

2) $f(x^{-1}) = f(x)^{-1}$

Proof

1) $f(e_G \cdot g) = f(g)$

$$f(e_G \cdot g) = f(e_G) \cdot f(g)$$

$$\Rightarrow f(e_G) \cdot f(g) = f(g)$$

this means $f(e_G)$ is a left identity in H
a similar argument shows $f(e_G)$ is right inverse.

$$\text{so } f(e_G) = e_H$$

$$2) \quad f(x x^{-1}) = f(e_G) = e_H$$

$$f(x x^{-1}) = f(x) f(x^{-1})$$

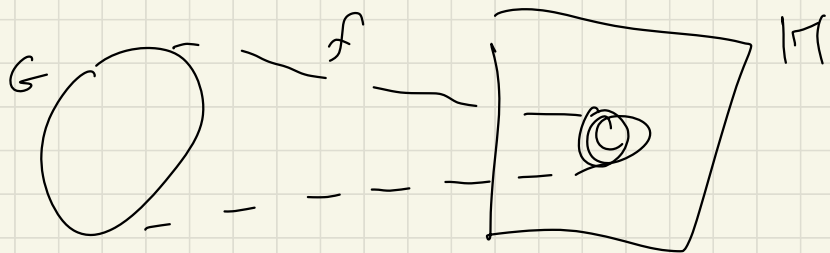
$$\text{so} \quad f(x) \cdot f(x^{-1}) = e_H$$

this means that $f(x^{-1})$ is a right inverse for $f(x)$
 a similar argument shows $f(x^{-1})$ is a left inverse for $f(x)$

$$f(x)^{-1} = f(x^{-1}) \quad \square$$

Def Let $f: G \rightarrow H$ be a hom. The set

$$\text{im}(f) = f(G) = \{f(x) : x \in G\} \subseteq H$$



is called the image of G in H under f .

claim 1) $\text{im}(f) \leq H$

$$2) f(K) = \{f(x) : x \in K\} \leq H \quad \text{for } K \leq G.$$

claim 1) $\text{im}(f) \leq H$

2) $f(K) = \{ f(x) : x \in K \} \leq H$ for $K \leq G$.

pf 2) identity $K \leq G$ so $e_G \in K$

$f(e_G) = e_H$ so $e_H \in f(K)$

closed under mult Let $f(x), f(y) \in f(K)$

$f(x) \cdot f(y) = f(xy)$ but $x, y \in K$

so $x \cdot y \in K$ so $f(xy) \in f(K)$

so $f(x) \cdot f(y) \in f(K)$

closed under inverse

$$\text{let } f(x) \in f(K)$$

$$\text{so } x \in K \Rightarrow x^{-1} \in K \Rightarrow f(x^{-1}) \in f(K)$$

$$\text{but } f(x^{-1}) = f(x)^{-1} \quad \text{so } f(x)^{-1} \in f(K)$$

Def: Let $f: G \rightarrow H$ be a ^{group} hom. □

$$\text{the set } \ker(f) = \{ x \in G : f(x) = e_H \}$$

is called the kernel of f .

3 claims from prop 12.7

Claim $\ker(f) \leq G$

Pf identity $f(e_G) = e_H$ so $e_G \in \ker(f)$

closed under mult Let $a, b \in \ker(f)$

$$f(a \cdot b) = f(a) \cdot f(b) = e_H \cdot e_H = e_H \quad \checkmark$$

closed under inverse Let $a \in \ker(f)$

$$f(a) = e_H$$

$$\Rightarrow f(a)^{-1} = e_H^{-1} = e_H$$

$$= f(a^{-1}) = e_H \Rightarrow a^{-1} \in \ker(f). \quad \square$$

Claim Let $f: G \rightarrow H$ be a group homomorphism, then
 f is one to one if and only if $\ker(f) = \{e_G\}$

Pf if f is one to one

Because f is a hom. $f(e_G) = e_H$

but because f is one to one $|\ker(f)| = 1$

$$\Rightarrow \{e_G\} = \ker(f)$$

if $\ker f = \{e_G\}$

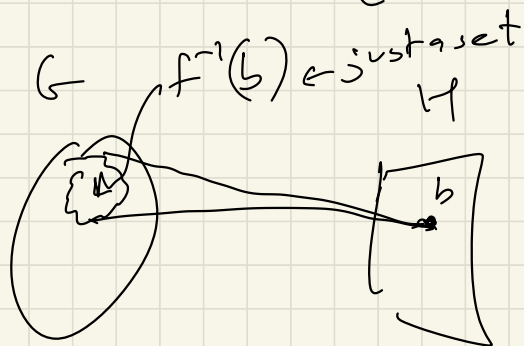
Let $f(a) = f(b)$ need to show is that $a = b$.

$$f(a) = f(b) \Rightarrow f(a) \cdot f(b)^{-1} = e_H \Rightarrow f(a \cdot b^{-1}) = e_H$$

$$\Rightarrow a \cdot b^{-1} \in \ker(f) \Rightarrow a \cdot b^{-1} = e_G \Rightarrow a = b \quad \square$$

Claim If $b = f(a)$ then $f^{-1}(b) = \{x \in G : f(x) = b\}$.
 preimage of b in G

This set, $f^{-1}(b)$ is the unique coset of
 $a \ker(f)$ in $G / \ker f$ containing a .



set of all cosets of $\ker f$
 in G

The claim is saying: if
 $b = f(a)$ then
 $f^{-1}(b) = a \ker(f)$

pf

$$x \in f^{-1}(b) \Leftrightarrow f(x) = b \quad (\text{def of } f^{-1})$$

$$\Leftrightarrow f(x) = f(g) \quad (\text{substitution})$$

$$\Leftrightarrow f(g)^{-1} f(x) = e_H \quad (\text{mult by } f(g)^{-1} \text{ on left})$$

$$\Leftrightarrow f(g^{-1}x) = e_H$$

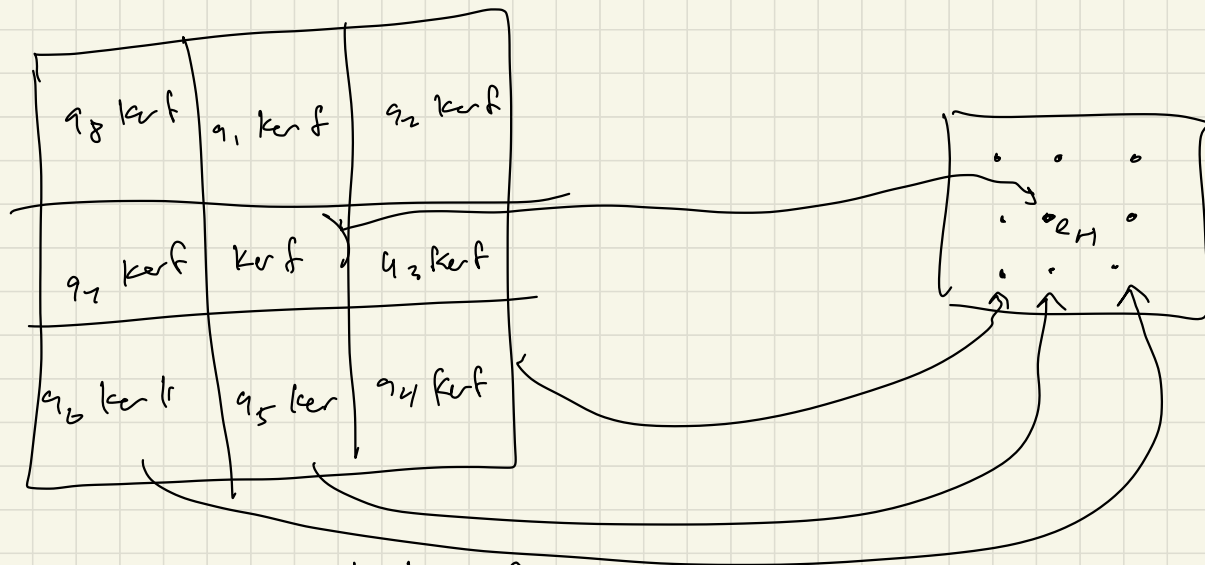
$$\Leftrightarrow g^{-1}x \in \text{Ker}(f)$$

$$\Leftrightarrow x \in g \text{Ker}(f)$$

$$\text{So } f^{-1}(b) = g \text{Ker}(f)$$

□

$$G \xrightarrow{f} H$$



cosets partition G

$$G / \ker f = \{ \ker f, g_1 \ker f, g_2 \ker f, g_3 \ker f, \dots, g_8 \ker f \}$$

leading question: How does multiplication in G translate to mult. in $G / \ker f$?